

Project 4

TITLE: Edge Computing Smart Home Safety System

OBJECTIVE:

To develop a smart home safety system using motion detection and emergency override logic.

COMPONENTS USED:

- * ESP32
- * PIR Motion Sensor
- * Potentiometer (Gas Sensor Simulation)
- * Red LED
- * Green LED
- * Buzzer
- * Wokwi Simulator

SOFTWARE USED:

- * Arduino IDE
- * Wokwi Simulator

WORKING PRINCIPLE:

The PIR sensor detects motion and activates smart lighting. The gas sensor continuously monitors gas levels. If gas concentration exceeds a threshold, emergency mode overrides all operations and activates alarms.

FEATURES:

- * Motion detection
- * Hardware interrupt logic
- * Emergency override system
- * Gas leak monitoring
- * Alarm and visual alerts

RESULT:

The system successfully detected motion and activated emergency alerts during simulated gas leaks.

CONCLUSION:

This project demonstrated interrupt handling, edge computing, and safety-critical embedded system design.